

SCOPE OF WORK:

1. Perform a comprehensive inspection of boilers for both safety and general condition and life expectancy.

Perform a comprehensive internal and external inspection prior to the start of project activities.

2. The Boiler Useful Life Studies and Condition Assessment to be used for establishing the engineering needed to determine the useful life expectancy and to establish a repair or maintenance scope of work that may be required for these boilers or tanks.

3. The inspection of the boiler scaling to be performed by a nondestructive testing method such as Ultrasonic.

The inspection for the Loss of tube thickness shall also be performed by a non-destructive testing method such as IRIS/Submerged UT (Internal Rotation I Inspection System).

The Tube sheets, shells and Furnace Tube shall be tested to determine if any of these areas may have been compromised by overheating.

4. Once the relevant information is gathered from the on-site inspection, the data can be compared to original construction data in most cases.

Often the National Board of Boiler and Pressure Vessels database can supply original manufacturers' data sheets, or the information may be provided by contractors Boiler Useful Life Study database for applicable boilers.

5. All of the inspection activities will be conducted by a National Board-Certified Boiler & Pressure Vessel Inspector.

All of the testing activities will be conducted by an American Society of Nondestructive Testing [ASNT] Certified Level II or Level III TechnicianInspection Requirements for Boiler

A. Consultant shall develop a detailed list of information required at least two weeks prior to the inspection to allow the VA to compile the information.

B. Consultant shall develop a detailed schedule of the inspection in consultation with the VA POC to ensure availability of equipment as needed during the execution of the project.

C. Visually inspect the following items and document the locations with a sketch or pictures. Any adverse findings must be documented with pictures and notes in detail. All data shall be included in the final report as supporting documentation.

1) Review the boiler/s history including operating conditions, date of last inspection, current inspection certificate, ASME stamping, National Board registration number, materials of construction, extent of post weld heat treatment, history of repairs and modifications.

2) All welded and threaded pipe connections to the boiler shell or drum for signs of leakage or defects in the welded areas. Defects must be documented as to location with a sketch or picture and be non-destructively tested.

- 3) Boiler support structure for proper alignment, evidence of sagging, shifting or misalignment.
- 4) Boiler casings for signs of warping, misalignment and overheating.
- 5) All furnace tubes and all accessible convection tubes for alignment, evidence of corrosion, erosion, bulges, sagging, defective or deteriorated welds.
- 6) Exposed surfaces of steam and mud drum in furnace, particularly ligament areas between tubes, for evidence of corrosion, cracking, scale buildup, other defects.
- 7) Refractory for cracking, shifting, sagging, bulging or general deterioration.
- 8) Internal areas of steam drum and mud drum (inside boiler shell in fire tube boilers).
 - a. Welded outlet connections.
 - b. Surface of drums for pitting or surface discontinuities.
 - c. Ligament areas between tube holes for cracking.
 - d. Each tube end and internal surface of representative number of tubes for pitting, erosion, corrosion and excessive scale buildup. Utilize fiber optic inspection equipment to inspect interior of tubes.
 - e. Clean and inspect shell or tube sheet between tube holes for cracking.